PRODUCT DATA SHEET

# SPEEDIVALVE

edwardsvacuum.com

**Edwards SP Speedivalves are diaphragm sealed in-line, isolation valves.**
The construction consists essentially of a flexible elastomeric diaphragm which is sealed onto a lightly polished seat by a screw thread mechanism. The mechanism is isolated from the system by the diaphragm resulting in an extremely rugged and 'dirty' system tolerant valve.
The valve terminates in NW flanges and can be pipeline supported or panel mounted (SP40K is pipeline supported only).



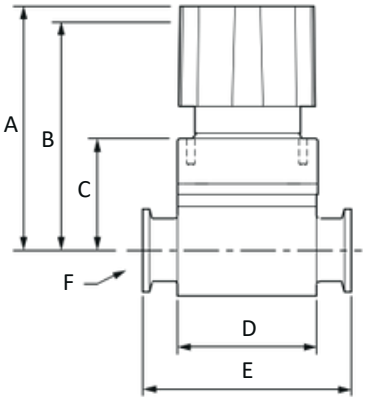
## FEATURES AND BENEFITS

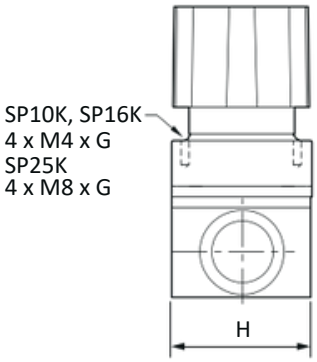
- Easy to operate with visual indication of valve open (SP10K to SP25K).
- Leak tight to better than 1 x $10^{-6}$ mbar $ls^{-1}$ (1 x $10^{-7}$ Torr $ls^{-1}$).
- Diaphragm completely isolates mechanism from vacuum system.
- Extremely rugged and 'dirty' system tolerant.
- Will withstand 9 bar overpressure (SP10K to SP40K, with co-seal).

## TECHNICAL DATA

| Speedivalve | |
|---|---|
| **Construction materials** | |
| Body | Aluminium alloy to BS1490 |
| Hand wheel and bonnet | Glass reinforced plastics |
| Diaphragms | Nitrile or Fluoroelastomer |
| **Leak rate (1 bar / 14.5 psi differential)** | |
| Valve (overall and across seat) | 1 x $10^{-6}$ mbar $ls^{-1}$ (1 x $10^{-7}$ Torr $ls^{-1}$) |
| Coupling | 1 x $10^{-7}$ mbar $ls^{-1}$ (1 x $10^{-8}$ Torr $ls^{-1}$) |
| **Molecular conductance** | |
| SP10K, SP16K | 1.7 $ls^{-1}$ |
| SP25K | 9.0 $ls^{-1}$ |
| SP40K | 23.3 $ls^{-1}$ |
| **Pressure rating using co-seal** | 9 bar / 131 psi |
| **Ambient operating range** | 0 to 40˚C |
| **Ambient storage range** | -10 to 40˚C |
| **Panel thickness** | 3 mm / 0.117 in maximum |
| **Weight** | |
| SP10K | 230 g / 8.1 oz |
| SP16K | 240 g / 8.4 oz |
| SP25K | 760 g / 26.6 oz |
| SP40K | 2300 g / 80.5 oz |
| **Baking temperature** | 60˚C |

## DIMENSIONS




| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| **Model** | **Open** | **Closed** | | | | | | |
| SP10K | 71 | 64 | 33.5 | 42 | 60 | NW10 | 8 | 43 |
| SP16K | 71 | 64 | 33.5 | 42 | 80 | NW16 | 8 | 43 |
| SP25K | 123 | 111 | 51 | 67 | 100 | NW25 | 12 | 72 |
| SP40K | 130 | - | - | 105 | 130 | NW40 | - | 96 |

## ORDERING INFORMATION

| Description | Order No. |
|---|---|
| SP10K, Nitrile Diaphragm | C33105000 |
| SP10K, Fluoroelastomer Diaphragm | C33155000 |
| SP16K, Nitrile Diaphragm | C33205000 |
| SP16K, Fluoroelastomer Diaphragm | C33255000 |
| SP25K, Nitrile Diaphragm | C33305000 |
| SP25K, Fluoroelastomer Diaphram | C33355000 |
| SP40K, Nitrile Diaphragm | C33405000 |
| SP40K, Fluoroelastomer Diaphragm | C33455000 |

| Description | Order No. |
|---|---|
| **Replacement Fluoroelastomer Diaphragms** | |
| Fluoroelastomer Diaphragm for SP10/16 | C33155800 |
| Fluoroelastomer Diaphragm for SP25 | C33355800 |
| Fluoroelastomer Diaphragm for SP40 | C33455800 |
| **Replacement Nitrile Diaphragms** | |
| Nitrile Diaphragm for SP10/16 | C33105800 |
| Nitrile Diaphragm for SP25 | C33305800 |
| Nitrile Diaphragm for SP40 | C33405800 |

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